

LU Decomposition

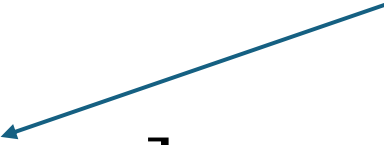
Prepared By: Tasnuva Binte Rahman

LU Decomposition/Factorization

We know, the system of linear equation can be expressed in the form: $Ax = b$(1)

A can be factorized into two triangular matrix L and U such that

$$A = LU \text{.....(2)}$$


$$\begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix}$$


$$\begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

LU Decomposition (contd.)

- Decompose a Non-Singular Square Matrix into LU form
- For a non-Singular Matrix A , $\det A \neq 0$
- L is the lower triangular matrix (Elements above diagonal are zero)
- U is the upper triangular matrix (Elements below diagonal are zero)

Why do we need LU Decomposition?

When the solution vector $\{B\}$ changes but the coefficient matrix $[A]$ is fixed, it is costly to repeat elimination method each time.

LU decomposition allows us to decompose $[A]$ into $[L]$ and $[U]$ *once*. Then, for each new solution vector, we only need to perform forward and backward substitution, while $[L]$ and $[U]$ remain fixed.

It is very useful for calculating the inverse matrix

Steps of LU Decomposition Method

$$\begin{array}{c} [A] \{x\} = \{b\} \\ \swarrow \quad \searrow \\ [L] [U] \{x\} = \{b\} \\ \quad \underbrace{\hspace{1.5cm}} \\ [L] \{d\} = \{b\} \dots\dots\dots(1) \text{ [Assuming, } [U]\{x\} = \{d\}] \end{array}$$

From eqn 1, we will get $\{d\}$ using **Forward Elimination**

$$[U] \{x\} = \{d\} \dots\dots\dots(2)$$

From eqn 2, we will get $\{x\}$ using **Backward Substitution**

Conventions

$$A = L U$$

For a unique LU decomposition, one triangular matrix is set to have unit diagonals.

Do-Little Method: L has 1s on the diagonal

Crout's Method: U has 1s on the diagonal

We will be using Do-Little Method in this slide.

In Crout's Method, the whole process is same except for U has 1s on the diagonal

Do-Little Decomposition Steps

Step1:LU Factorization

Firstly, coefficient matrix A is decomposed to L and U.

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

Step2: Forward Substitution

Now, the L and solution vector b is used to generate intermediate vector d.

$$\begin{aligned} [L]\{d\} &= \{b\} \\ \Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix} &= \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \end{aligned}$$

Do-Little Decomposition Steps (contd.)

Step3: Backward Substitution

The result from forward substitution is used to solve the unknown vector $\{x\}$ by substituting d in $[U]\{x\} = \{d\}$

$$\Rightarrow \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}$$

LU Factorization

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

Multiplying L and U we get,

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ l_{21}u_{11} & l_{21}u_{12} + u_{22} & l_{21}u_{13} + u_{23} \\ l_{31}u_{11} & l_{31}u_{12} + l_{32}u_{22} & l_{31}u_{13} + l_{32}u_{23} + u_{33} \end{bmatrix}$$

Equating the coefficients of A and L.U we get,

$$\begin{array}{lll} u_{11} = a_{11} & u_{12} = a_{12} & u_{13} = a_{13} \\ l_{21} = \frac{a_{21}}{u_{11}} & u_{22} = a_{22} - l_{21}u_{12} & \end{array}$$

[Just check which one is your unknown, keep it on left and put the rest on RHS]

LU Factorization (contd.)

Equating the coefficients of A and L.U we get,

$$u_{11} = a_{11}$$

$$u_{12} = a_{12}$$

$$u_{13} = a_{13}$$

$$l_{21} = \frac{a_{21}}{u_{11}}$$

$$u_{22} = a_{22} - l_{21}u_{12}$$

$$u_{23} = a_{23} - l_{21}u_{13}$$

$$l_{31} = \frac{a_{31}}{u_{11}}$$

$$l_{32} = \frac{a_{32} - l_{31}u_{12}}{u_{22}}$$

$$u_{33} = a_{33} - l_{31}u_{13} - l_{32}u_{23}$$

Now, as all entries of U and L matrices are known, we can use them to perform step 2 and 3.

Example of LU Factorization of A

$$A = \begin{bmatrix} 1 & 2 & 4 \\ 3 & 8 & 14 \\ 2 & 6 & 13 \end{bmatrix} \quad x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \quad d = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}$$

We will try to find L and U now

$$\begin{bmatrix} 1 & 2 & 4 \\ 3 & 8 & 14 \\ 2 & 6 & 13 \end{bmatrix} = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ l_{21}u_{11} & l_{21}u_{12} + u_{22} & l_{21}u_{13} + u_{23} \\ l_{31}u_{11} & l_{31}u_{12} + l_{32}u_{22} & l_{31}u_{13} + l_{32}u_{23} + u_{33} \end{bmatrix}$$

$$u_{11} = 1, u_{12} = 2, u_{13} = 4, l_{21} = \frac{a_{21}}{u_{11}} = \frac{3}{1} = 3,$$

$$u_{22} = a_{22} - l_{21}u_{12} = 8 - (3 \times 2) = 2, u_{23} = a_{23} - l_{21}u_{13} = (14 - 3 \times 4) = 2$$

$$l_{31} = \frac{a_{31}}{u_{11}} = \frac{2}{1} = 2, l_{32} = \frac{a_{32} - l_{31}u_{12}}{u_{22}} = \frac{6 - (2 \times 2)}{2} = 1,$$

$$u_{33} = a_{33} - l_{31}u_{13} - l_{32}u_{23} = 13 - (2 \times 4) - (1 \times 2) = 3$$

Example of LU Factorization of A

$$A = \begin{bmatrix} 1 & 2 & 4 \\ 3 & 8 & 14 \\ 2 & 6 & 13 \end{bmatrix} \quad x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \quad d = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}$$

We will try to find L and U now

$$\begin{bmatrix} 1 & 2 & 4 \\ 3 & 8 & 14 \\ 2 & 6 & 13 \end{bmatrix} = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ l_{21}u_{11} & l_{21}u_{12} + u_{22} & l_{21}u_{13} + u_{23} \\ l_{31}u_{11} & l_{31}u_{12} + l_{32}u_{22} & l_{31}u_{13} + l_{32}u_{23} + u_{33} \end{bmatrix}$$

$$u_{11} = 1, u_{12} = 2, u_{13} = 4, l_{21} = \frac{a_{21}}{u_{11}} = \frac{3}{1} = 3,$$

$$u_{22} = a_{22} - l_{21}u_{12} = 8 - (3 \times 2) = 2, u_{23} = a_{23} - l_{21}u_{13} = (14 - 3 \times 4) = 2$$

$$l_{31} = \frac{a_{31}}{u_{11}} = \frac{2}{1} = 2, l_{32} = \frac{a_{32} - l_{31}u_{12}}{u_{22}} = \frac{6 - (2 \times 2)}{2} = 1,$$

$$u_{33} = a_{33} - l_{31}u_{13} - l_{32}u_{23} = 13 - (2 \times 4) - (1 \times 2) = 3$$

Example of LU Factorization of A (Contd.)

$$\text{So, } \begin{bmatrix} 1 & 2 & 4 \\ 3 & 8 & 14 \\ 2 & 6 & 13 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 2 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 4 \\ 0 & 2 & 3 \\ 0 & 0 & 3 \end{bmatrix}$$

Now that we have seen how to decompose A into LU, we will now see how to solve a system of linear equation.

Example 1

Solve the following system using LU decomposition.

$$x_1 + x_2 + x_3 = 1$$

$$4x_1 + 3x_2 - x_3 = 6$$

$$3x_1 + 5x_2 + 3x_3 = 4$$

Solution:

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 4 & 3 & -1 \\ 3 & 5 & 3 \end{bmatrix}, \quad x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 6 \\ 4 \end{bmatrix}$$

Example 1 (contd.)

Step 1: LU decomposition of A

We will try to find L and U now

$$\begin{bmatrix} 1 & 1 & 1 \\ 4 & 3 & -1 \\ 3 & 5 & 3 \end{bmatrix} = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ l_{21}u_{11} & l_{21}u_{12} + u_{22} & l_{21}u_{13} + u_{23} \\ l_{31}u_{11} & l_{31}u_{12} + l_{32}u_{22} & l_{31}u_{13} + l_{32}u_{23} + u_{33} \end{bmatrix}$$

$$u_{11} = 1, u_{12} = 1, u_{13} = 1, l_{21} = \frac{a_{21}}{u_{11}} = \frac{4}{1} = 4,$$

$$u_{22} = a_{22} - l_{21}u_{12} = 3 - (4 \times 1) = -1, u_{23} = a_{23} - l_{21}u_{13} = (-1 - 4 \times 1) = -5$$

$$l_{31} = \frac{a_{31}}{u_{11}} = \frac{3}{1} = 3, l_{32} = \frac{a_{32} - l_{31}u_{12}}{u_{22}} = \frac{5 - (3 \times 1)}{-1} = -2,$$

$$u_{33} = a_{33} - l_{31}u_{13} - l_{32}u_{23} = 3 - (3 \times 1) - (-2 \times -5) = 3 - 3 - 10 = -10$$

Example 1 (contd.)

$$L = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix}, U = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

By putting the values to L and U we get,

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 3 & -2 & 1 \end{bmatrix}, U = \begin{bmatrix} 1 & 1 & 1 \\ 0 & -1 & -5 \\ 0 & 0 & -10 \end{bmatrix}$$

Step 2: Applying Forward Substitution considering

$$[L]\{d\} = \{b\}$$

Example 1 (contd.)

$$\Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 3 & -2 & 1 \end{bmatrix} \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 6 \\ 4 \end{bmatrix}$$

$$\mathbf{d_1 = 1}$$

$$4d_1 + d_2 = 6$$

$$\Rightarrow \mathbf{d_2 = 6 - 4d_1 = 6 - 4 = 2}$$

$$3d_1 - 2d_2 + d_3 = 4$$

$$\Rightarrow \mathbf{d_3 = 4 - 3d_1 + 2d_2 = 4 - 3 + 4 = 5}$$

Example 1 (contd.)

Step 3: Applying Backward Substitution considering $[U]\{x\} = \{d\}$

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & -1 & -5 \\ 0 & 0 & -10 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 5 \end{bmatrix}$$

$$x_3 = \frac{5}{-10} = \frac{-1}{2}$$

$$-x_2 - 5x_3 = 2$$

$$\Rightarrow x_2 = -5x_3 - 2 = -\frac{-5}{2} - 2 = \frac{1}{2}$$

$$x_1 + x_2 + x_3 = 1$$

$$\Rightarrow x_1 = 1 - x_2 - x_3 = 1 - \frac{1}{2} + \frac{1}{2} = 1$$

LU Decomposition using Gauss Elimination

Solve the following system using LU decomposition.

$$3x_1 + 2x_2 + x_3 = 10$$

$$2x_1 + 3x_2 + 2x_3 = 14$$

$$x_1 + 2x_2 + 3x_3 = 14$$

Solution:

$$A = \begin{bmatrix} 3 & 2 & 1 \\ 2 & 3 & 2 \\ 1 & 2 & 3 \end{bmatrix}, \quad x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \quad b = \begin{bmatrix} 10 \\ 14 \\ 14 \end{bmatrix}$$

Note: Basically, when we apply Gauss Elimination, we get upper matrix, we will use that technique to get L and U. Rest of the process is as we have done before

LU Decomposition using Gauss Elimination (contd.)

Step 1: Eliminate x_1 from R2

$$R1: 3x_1 + 2x_2 + x_3 = 10..(1) \text{ Pivot Row}$$

$$R2: 2x_1 + 3x_2 + 2x_3 = 14..(2) \text{ Elimination Row}$$

$$R3: x_1 + 2x_2 + 3x_3 = 14..(3)$$

From (2) – (1) * *pivot* $\Rightarrow$

where, **pivot** = $\frac{2}{3}$

$$\begin{array}{r} 2x_1 + 3x_2 + 2x_3 = 14 \\ -2x_1 - \frac{4}{3}x_2 - \frac{2}{3}x_3 = \frac{-20}{3} \\ \hline \frac{5}{3}x_2 + \frac{4}{3}x_3 = \frac{22}{3} \dots\dots (2') \end{array}$$

LU Decomposition using Gauss Elimination (contd.)

Step 2: Eliminate x_1 from R3

$$R1: 3x_1 + 2x_2 + x_3 = 10..(1) \text{ Pivot Row}$$

$$R2: 2x_1 + 3x_2 + 2x_3 = 14..(2)$$

$$R3: x_1 + 2x_2 + 3x_3 = 14..(3) \text{ Elimination Row}$$

From (3) – (1) * *pivot* $\Rightarrow$

where, **pivot** = $\frac{1}{3}$

$$\begin{array}{rcl} x_1 + 2x_2 + 3x_3 & = & 14 \\ -x_1 - \frac{2}{3}x_2 - \frac{1}{3}x_3 & = & \frac{-10}{3} \\ \hline \frac{4}{3}x_2 + \frac{8}{3}x_3 & = & \frac{32}{3} \dots\dots (2') \end{array}$$

LU Decomposition using Gauss Elimination (contd.)

Step 3: Eliminate x_2 from R3

$$R1: 3x_1 + 2x_2 + x_3 = 10 \dots (1)$$

$$R2': \quad \frac{5}{3}x_2 + \frac{4}{3}x_3 = \frac{22}{3} \dots \dots (2') \text{ Pivot Row}$$

$$R3': \quad \frac{4}{3}x_2 + \frac{8}{3}x_3 = \frac{32}{3} \dots \dots (3') \text{ Elimination Row}$$

From $(3') - (2') * pivot \Rightarrow$

where, **pivot** = $\frac{4}{5}$

$$\frac{4}{3}x_2 + \frac{8}{3}x_3 = \frac{32}{3}$$

$$-\frac{4}{3}x_2 - \frac{16}{15}x_3 = -\frac{88}{15}$$

$$\frac{8}{5}x_3 = \frac{24}{5} \dots \dots (3'')$$

LU Decomposition using Gauss Elimination (contd.)

Now the system becomes,

$$R1: \quad 3x_1 + 2x_2 + x_3 = 10 \dots (1)$$

$$R2': \quad \frac{5}{3}x_2 - \frac{4}{3}x_3 = \frac{22}{3} \dots \dots (2')$$

$$R3'': \quad \frac{8}{5}x_3 = \frac{24}{5} \dots \dots (3'')$$

$$\text{So, } U = \begin{bmatrix} 3 & 2 & 1 \\ 0 & \frac{5}{3} & \frac{4}{3} \\ 0 & 0 & \frac{8}{5} \end{bmatrix}, \quad L = \begin{bmatrix} 1 & 0 & 0 \\ \frac{2}{3} & 1 & 0 \\ \frac{1}{3} & \frac{4}{5} & 1 \end{bmatrix}$$

Note: Think of L as originally an identity matrix. When you are eliminating terms using a pivot, the corresponding multipliers are placed in L, resulting in the L matrix.

LU Decomposition using Gauss Elimination (contd.)

Here,

$$L = \begin{bmatrix} 1 & 0 & 0 \\ \frac{2}{3} & 1 & 0 \\ \frac{1}{3} & \frac{4}{5} & 1 \end{bmatrix}; l_{21} = \frac{a_{21}}{u_{11}}, l_{31} = \frac{a_{31}}{u_{11}}, l_{32} = \frac{a'_{32}}{a'_{22}}$$

Now performing forward substitution using $[L]\{d\} = \{b\}$

$$\begin{bmatrix} 1 & 0 & 0 \\ \frac{2}{3} & 1 & 0 \\ \frac{1}{3} & \frac{4}{5} & 1 \end{bmatrix} \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix} = \begin{bmatrix} 10 \\ 14 \\ 14 \end{bmatrix}$$

$$d_1 = 10$$

$$\frac{2}{3}d_1 + d_2 = 14$$

$$\Rightarrow d_2 = 14 - \frac{2}{3}d_1 = 14 - \frac{20}{3} = \frac{22}{3}$$

LU Decomposition using Gauss Elimination (contd.)

$$\frac{1}{3}d_1 + \frac{4}{5}d_2 + d_3 = 14$$

$$\Rightarrow \mathbf{d}_3 = 14 - \frac{1}{3}d_1 - \frac{4}{5}d_2 = 14 - \frac{10}{3} - \frac{88}{15} = \frac{24}{5}$$

Now, applying backward substitution by $[U]\{x\} = \{d\}$ we get,

$$U = \begin{bmatrix} 3 & 2 & 1 \\ 0 & \frac{5}{3} & \frac{4}{3} \\ 0 & 0 & \frac{8}{5} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 10 \\ 22/3 \\ 24/5 \end{bmatrix}$$

$$x_3 = 3$$

$$\frac{5}{3}x_2 + \frac{4}{3}x_3 = \frac{22}{3}$$

LU Decomposition using Gauss Elimination (contd.)

$$\Rightarrow x_2 = \frac{\frac{22}{3} - \frac{4}{3}x_3}{\frac{5}{3}} = \frac{\frac{22}{3} - \frac{4}{3} \cdot 3}{\frac{5}{3}} = 2$$

$$3x_1 + 2x_2 + x_3 = 10$$

$$\Rightarrow x_1 = \frac{10 - 2x_2 - x_3}{3} = \frac{10 - 2 \cdot 2 - 3}{3} = 1$$

Final Solution, $x_1 = 1, x_2 = 2, x_3 = 3$